

Solve-and-Graph Relay

Unit 7: Equation Detective Agency · Lesson 7-6 · EduWonderLab

UNIT 7 · LESSON 7-6 · TEACHER NOTES

Standard 6.EE.5

FORMAT Team relay

TIME 25–35 min

GROUPING Teams of 3–4

TOPIC Solve and Graph Inequalities

STANDARD 6.EE.5

PREP LEVEL Light — print relay cards and number-line strips

Learning goal *I can solve a one-step inequality, check the boundary, and graph the solution.*

Teacher setup

- Print one relay set per team. Each card is solved, then graphed.
- Teammates pass the work along: solve, check the boundary, graph, verify.
- The checklist confirms each step before the card is done.

Materials

Relay cards, Number line strips, Verification checklist, Pencil.

Differentiation & language support

- Verification checklist: solution, circle, direction (SPED).
- Sentence stem: “The graph shows all numbers that ____.”
- ESOL: test one value to confirm the solution set.

Key vocabulary (English / Español):

Term	Español	What it means
solution set	conjunto solución	all the values that make it true
boundary	límite	the number where the solution starts

Quick teacher check: *Check the solve step and circle type on Card 4 ($x \div 3 \geq 2$).*

Solve-and-Graph Relay

Unit 7: Equation Detective Agency · Lesson 7-6 · EduWonderLab

Name: _____ Date: _____ Class: _____

Your goal: *I can solve a one-step inequality, check the boundary, and graph the solution.*

What you need

Relay cards, Number line strips.

How to play

1. Teammate 1 solves the inequality.
2. Teammate 2 checks the boundary value.
3. Teammate 3 graphs the solution set.
4. Teammate 4 verifies with the checklist, then pass to the next card.

Relay Cards

Solve the inequality, then graph the solution. Check by testing a value.

1. $x + 3 > 7$	2. $x - 2 \leq 5$
3. $2x < 10$	4. $x \div 3 \geq 2$
5. $x + 5 \leq 9$	6. $x - 4 > 1$
7. $4x \geq 12$	8. $x \div 2 < 4$

Record your work (solve, then graph):

Explain your thinking

Explain why an inequality has many solutions instead of just one.

Sentence frame: *The graph shows all numbers that ____.*

★ **Challenge:** For Card 1 ($x > 4$), name three numbers in the solution and one number that is not.

ANSWER KEY

Teacher copy — please do not distribute to students.

UNIT 7 • LESSON 7-6 • Solve-and-Graph Relay

Standard 6.EE.5

Answers

1. $x + 3 > 7$ Answer: $x > 4$ — open circle at 4, shade right	2. $x - 2 \leq 5$ Answer: $x \leq 7$ — closed circle at 7, shade left
3. $2x < 10$ Answer: $x < 5$ — open circle at 5, shade left	4. $x \div 3 \geq 2$ Answer: $x \geq 6$ — closed circle at 6, shade right
5. $x + 5 \leq 9$ Answer: $x \leq 4$ — closed circle at 4, shade left	6. $x - 4 > 1$ Answer: $x > 5$ — open circle at 5, shade right
7. $4x \geq 12$ Answer: $x \geq 3$ — closed circle at 3, shade right	8. $x \div 2 < 4$ Answer: $x < 8$ — open circle at 8, shade left

Teaching notes & look-fors

- Solve like an equation, then graph the solution set.
- Card 4: $x \div 3 \geq 2 \rightarrow x \geq 6$ (closed circle, shade right).